COMP232 CyberSecurity

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Organisation

- Resources & lectures will be available on CANVAS
- 2 Hours Lecture Session/week
- 1 Hour Lab Session/week

Aims

1. To provide students with understanding of the main problems in security, confidentiality and privacy in computers and in networks, and the reasons for their importance.

2. To enable students to understand the main approaches adopted for their solution and/or mitigation, together with the strengths and weaknesses of each of these approaches.

3. To develop knowledge and skills in practical applications of available security solutions.

4. To introduce students to theoretical foundations of cybersecurity and attract their attention to the open problems requiring further research.

Textbooks

Main:

- Richard R. Brooks, **Introduction to Computer and Network Security, Navigating Shades of Gray**, CRC Press, Taylor and Francis Group, 2014 (and later editions). CNS

Additional:

- William Stallings , **Network Security Essentials: Applications and Standards**. Prentice Hall, 2000 (and later editions). NSE
- A. Menezes, P. van Oorschot, and S. Vanstone, **Handbook of Applied Cryptography**, CRC Press, 1996. Available online at http://www.cacr.math.uwaterloo.ca/hac free for personal use;

Assessment

- Cybersecurity is both conceptual and practical
- This module is assessed in two complementary ways

- Two-hour written exam (70%)
- Focus on understanding and reasoning about cybersecurity concepts, algorithms, protocols, and attacks. Marks are awarded for clear explanations and sound reasoning. The mock exam is representative of the style and level expected.
- Coursework - short technical reports (30%)
- The coursework is divided into two assignments (15% each), closely linked to the lab sessions. The assignments focus on analysing and evaluating security mechanisms in practice, interpreting results, and communicating findings clearly.

Keep up with lectures, actively engage with labs, and start coursework early, there should be no surprises in either component.

More Information

- Lab sessions:

  - start at week 2, see your personal timetable

  - Demonstrators are your main point of call

- Assignments Deadlines (Canvas will be kept updated):

  1 - 13 MAR 2026 (17:00)

  2 - 1 May 2026 (17:00)

- Feedback:

  Feedback is most useful when it’s timely

  if something is unclear or not working for you, let me know early so it can be fixed during the module.

Cybersecurity: What is it?

Cyberspace:

is an electronic medium used to form a global computer network(s) to facilitate online communication

Security:

A condition that results from **establishment** and **maintenance** of **protective measures** that ensure a state of inviolability from **hostile acts** or influences

Security in Cyberspace

In the modern world there are various ways in which hostile act and influences can be exercised.

Many of them are coming via ‘Cyberspace’, where in particular, unprecedented amount of data about individuals and organizations being collected, processed, analysed and possibly misused.

Important Concept

Privacy:

- Privacy is the ability of a person to control the availability of information about and exposure of themselves. It is related to being able to function in society anonymously… (from Wikipedia)

Not the same, but Interlinked with **Security**, as

- Availability of private information may itself constitute a hostile act
- Availability of (or acquiring) some information may be a precondition for some security attacks

What makes it interesting

- Computer Science: new opportunities/challenges
- Mathematics: non-trivial mathematics behind many solutions
- Physics: rise of quantum computation (Shor '94) and quantum cryptography
- Biology: DNA-analysis based authentication
- Technology: global networking, cloud computing
- Economical issues: costly cybersecurity
- Legal & Political Issues: is it legal to fight back? Political influence by interfering with elections, etc
- Social and moral aspects: shall we trade privacy for better security?

Costly Cybersecurity

- The worldwide information security market is forecast to reach **$170.4 billion** in 2022. [1]
- 62% of businesses experienced **phishing** and **social engineering** attacks in 2018. [2]
- 68% of business leaders feel their cybersecurity risks are increasing. [3]
- **Data breaches** exposed 4.1 billion records in the first half of 2019. [4]
- The top malicious **email attachment** types are .doc and .dot which make up 37%, the next highest is .exe at 19.5%. [5]

Global impact:

- The average cost of a malware attack on a company is $2.6 million. [1] Can be hundreds of millions of $.
- **$**3.9 million is the average cost of a data breach. [2]
- The financial services industry takes in the highest cost from cybercrime at an average of $18.3 million per company surveyed. [3]
- The industry with the highest number of attacks by ransomware is the healthcare industry. Attacks quadrupled in 2020. [4]

Jobs

- 82% of employers report a **shortage of cybersecurity skills**. [1]
- In 2021, 3.5 million unfilled cybersecurity jobs globally. [2]
- Since 2016, the demand for Data Protection Officers (DPOs) has skyrocketed and risen over 700%, due to the **GDPR demands**. [3]

What can hackers do with my information?

- **Sell your data to other hackers**
  - One way hackers profit from stolen data is selling it in masses to other criminals on the dark web. These collections can include millions of records of stolen data. The buyers can then use this data for their own criminal purposes.
- **Identity theft**
  - using the victim's credit card or taking loans in their name.
- **used to target phishing attacks and extortion**
  - With stolen personal information criminals can target victims with phishing attacks. In phishing scams victims are lured into giving information like credit card details willingly to criminals by masking the scam as something legit. If criminals get access to very sensitive information, they can also extort the victim.

$1,200

is all you are worth on the dark web

Hacked accounts of these popular brands and stolen personal info are for sale on the dark web. The Dark Web Market Price Index tracks their average sale price, showing fraudsters can buy up someone's entire online identity for just $1,200.

		Subtotal
Online Shopping	amazon, ebay, Costco Wholesale, Walmart, macy's	$164.65
Travel	UBER, Booking.com, Expedia, airbnb	$45.53
Entertainment	NETFLIX, Spotify	$28.59
Personal Finance	PayPal	$710.65
Social Media	facebook, LinkedIn	$10.21
Proof of Identity		$92.20
Communication	verizon, AT&T, T-Mobile	$72.17
Delivery	DHL, ups, FedEx Express	$15.59
Food Delivery	GRUBHUB, Pizza Hut	$12.80
Email	Aol., YAHOO! MAIL	$9.53
Dating	match.com, dating.com	$8.82

Source: Dark web market listings collected on 5-11 February, 2018. Markets monitored were Dream, Point and Wall Street Market. Prices collected in USD as displayed on listings.

TOPIOVPN

Your identity is a steal on the Dark Web.

Here are what the most common pieces of information sell for:

experian.

Social security number	$1
Online payment services login info (e.g. Paypal)	$20-$200
Credit or debit card (credit cards are more popular)	$5-$110
With CVV number	$5
With bank info	$15
Fullz info*	$30
Drivers license	$20
Loyalty accounts	$20
General non-financial institution logins	$1
Diplomas	$100-$400
Passports (US)	$1000-$2000
Subscription services	$1-$10
Medical records	$1-$1000**

Credit Card Details	Avg. Price USD (2021)
Hacked (Global) credit card details with CVV	**$35**
USA hacked credit card details with CVV	**$17**
UK hacked credit card details with CVV	**$20**
Canada hacked credit card details with CVV	**$28**
Australia hacked credit card details with CVV	**$30**
Israel hacked credit card details with CVV	**$65**
Spain hacked credit card details with CVV	**$40**
Japan hacked credit card details with CVV	**$40**

Product	Avg. Price USD (2020)	Avg. Price USD (2021)	YoY Diff
Cloned Mastercard with PIN	$15	$25	+$10
Cloned American Express with PIN	$35	$35	$0
Cloned VISA with PIN	$25	$25	$0
Credit card details, account balance up to $1,000	$12	$15	+$3
Credit card details, account balance up to $5,000	$20	$24	+$4
Stolen online banking logins, minimum $100 on account	$35	$40	+$5
Stolen online banking logins, minimum $2,000 on account	$65	$120	+$55
Walmart account with credit card attached	$10	$14	+$4

Computer security in Industrial software

- Stuxnet
- Computer worm discovered in June 2010
- It targets Siemens industrial software and equipment running on Microsoft Windows
- 60% of the infected computers were in Iran (August 2010) including controllers handling the centrifuges at Natanz **nuclear facilities**
- Was it a field test of a **cyber weapon**?

- Also, hacking Cars (2015), Medical equipments etc.

SolarWinds

- SolarWinds is a company that produces an extremely popular network management system – Credential management.
- Software based – manages all of the credentials on the network. Specializes in managing them for large organizations and enterprises
- They gained access to victims via trojanized updates to SolarWind's Orion IT monitoring and management system.
- 'SolarWinds.Orion.Core.BusinessLayer.dll' is a SolarWinds digitally signed component of the Orion software framework that contains a backdoor that communicates via HTTP to third-party servers.
- Initial infection started in 2019 with a dormant ‘do nothink’ commit to the main source code repository

Content of the course

✓ Security and Privacy Overview

- security attributes, authentication and authorization, access permission, social engineering, vulnerabilities and attacks.
- Passwords v. tokens v. biometrics, Multifactor authentication

✓ Cryptography

- symmetric encryption, public key encryption, hash functions, key exchange protocols, message confidentiality.
- steganography, homomorphic encryption
- quantum cryptography and computation
- zero-knowledge proofs, secure multiparty computations

✓ Security protocols

- key exchange, man-in-the-middle attack, formal methods for protocol analysis, model checking

Content

✓ Securing Networks:

- firewalls, intrusion detection and prevention systems.

✓ Data Aggregation, Privacy, and Anonymity

- Mix-networks, Crowds, Tor-network

✓ Attacks and defences:

- SQL Injection, SSH insertion, Malwares, Anti-virus methods

✓ legal and ethical issues related to security and privacy

✓ Applications of cryptographic algorithms

Reading

[CNS]:

Chapter 1

[NSE]:

Chapter 1, sections 1.1 –1.3